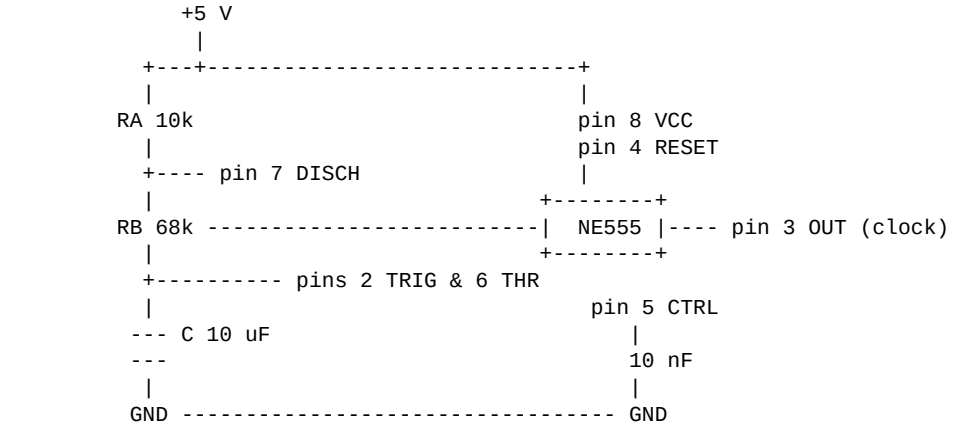
NE555 Astable Clock & 8-Bit ALU IC Datasheet Guide

For the ALUForge '26 8-bit ALU hardware project

1. NE555P in astable mode — approximately 1 Hz

Suggested timing parts: RA = 10 kΩ, RB = 68 kΩ, C = 10 µF. The calculated frequency is about 0.99 Hz (actual value varies with component tolerance and capacitor leakage).

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Pin 1 GND -> GND. Join pin 2 and pin 6.
RA: +5 V to pin 7. RB: pin 7 to pins 2/6.
C: pins 2/6 to GND. Pin 3 is the clock output.
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Frequency formula: f ≈ 1.44 / ((RA + 2RB) × C)

For RA = 10,000 Ω, RB = 68,000 Ω, C = 10 µF: f ≈ 1.44 / (146,000 × 0.000010) ≈ 0.99 Hz. Approximate duty cycle: (RA + RB)/(RA + 2RB) ≈ 53%.

Connections: Pin 1 → GND; pin 8 → +5 V; pin 4 RESET → +5 V; pins 2 and 6 tied together; pin 7 between RA and RB; pin 3 clock output; pin 5 CTRL → 10 nF capacitor to GND. Connect all grounds together. Add a 100 nF bypass capacitor between +5 V and GND close to the NE555.

Note: TI specifies the NE555 supply range as 4.5 V to 16 V, so regulated +5 V is suitable. Verify the clock edge and register requirements against the SN74LS173A datasheet before connecting it.

2. Official datasheets for all ICs in your list

Open each manufacturer's page and select the datasheet under Technical documentation / Data sheet.

IC and quantity	Purpose	Official datasheet page
SN74LS173AN — 4	4-bit register with 3-state outputs; two per 8-bit register.	TI SN74LS173A
SN74HCT245N — 2	8-bit bidirectional bus transceiver with tri-state outputs.	TI SN74HCT245
SN74LS283N — 2	4-bit binary full adder; two cascaded for 8-bit addition.	TI SN74LS283
SN74HC86N — 2	Quad XOR gates for conditional B inversion during subtraction.	TI SN74HC86
NE555P — 1	Timer IC used as an approximately 1 Hz astable clock.	TI NE555

3. Important design notes

• SN74LS173AN is a 4-bit register; four ICs can provide two 8-bit registers for operands A and B.

• A common add/subtract design uses A + (B XOR SUB) + SUB, where SUB = 0 selects addition and SUB = 1 selects subtraction. The least-significant carry-in must be connected to SUB.

• Set the SN74HCT245 DIR and OE controls carefully. Never enable multiple drivers onto the same bus at once, as this can cause bus contention.

• Check the datasheet for the exact part marking and package you own before wiring power or signals.